

Curriculum Vol 0 Architectural Scaffold v0.1 — Substrate Foundation

Keeper (curriculum-wide integration architect)

2026-05-21 Thursday morning

Curriculum Vol 0 Architectural Scaffold v0.1

Purpose

Vol 0 is the prerequisite for every other volume in the BST Physics Curriculum. It defines: - WHAT D_{IV}^5 is (the substrate object) - WHAT the 5 integers + N_{max} are (the substrate primaries) - HOW the substrate operates (commitment cycle, 4 zones, 3 scales) - WHY it's uniquely forced (Strong-Uniqueness Theorem) - HOW we read it (operators + trichotomy + tier discipline)

Every Vol 1-10 chapter cites Vol 0 for the ontology. Vol 0 cites itself.

This scaffold specifies the chapter STRUCTURE; Grace's parallel work specifies the chapter CONTENT outline. Combined they form Vol 0 v0.2.

Proposed chapter structure (10 chapters + 2 appendices)

Part I — The Substrate Object

Chapter 1: D_{IV}^5 — the Autogenic Proto-Geometry - $D_{IV}^5 = SO_0(5,2) / [SO(5) \times SO(2)]$ as Lie group quotient - Bergman geometry primer - Why this geometry (Strong-Uniqueness preview) - Reader exits with: clear mental model of the substrate's mathematical object - Believability anchor: "this is a specific manifold mathematicians have studied since Cartan"

Chapter 2: The Five Integers + N_{max} - rank = 2, $N_c = 3$, $n_C = 5$, $C_2 = 6$, $g = 7$, $N_{max} = 137$ - Per-integer "Why X" theorems (T1925, T1930, SP-31-39 consolidations) - The integers as substrate signatures - Reader exits with: concrete handle on the BST primaries - Provability anchor: 6 per-integer theorems registered as D-tier

Chapter 3: The Substrate Operating System - Substrate Working Process Principle (SWPP) — absorption / commitment / emission - Koons tick clock (T2405) - 4-Zone Commitment Cycle (T2420) - 3-Scale Substrate Operation (intra / inter-local / inter-distant) - Reed-Solomon coding on $GF(2^g)$ (K59) - Reader exits with: substrate as

active information-cycle, not static geometry

Part II — Substrate Symmetries and Structure

Chapter 4: Isotropy Group Structure - $SO(5) \times SO(2)$ factor decomposition - $SO(5)$ gives spacetime-side observables - $SO(2)$ gives internal-symmetry observables (charge, chirality) - Möbius involution gives parity - Full $SO_0(5,2)$ gives time-translation (Hamiltonian) - Reader exits with: organizing principle for the operator zoo

Chapter 5: Boundary Conditions (the 6+2 BC framework) - Substrate boundary conditions: 6 internal + 2 external - Per-BC theorems (SP-31-40, forthcoming) - Bulk-Boundary Two-Face Structure of integer-webs - Reader exits with: what closes substrate; what couples it to external

Chapter 6: The Integer Web Principle - Each BST primary as a “boundary” holding a web of structural relations - Casey vision Wednesday EOD (filed as Casey-named principle) - Cross-classification matrix (Grace 256-cell) - Reader exits with: integers as anchor-points for substrate’s combinatorial graph

Part III — Substrate Observables

Chapter 7: The Operator Zoo - Hilbert space specification (SP-31-1, forthcoming Lyra) - Operator inventory: Position (T2419) + Momentum (T2422) + Spin (T2421) + Angular Momentum (#247) + Bell-CHSH (K66, trace-level per Cal #17) + Charge ($SO(2)$ weight) + Chirality (twistor $SO(2)$) + Number (cycle-count) + Parity (Möbius) + Hamiltonian (pending K52a S24+) + Time (Koons tick) - Organizing principle: $SO(5) \times SO(2) \times$ Möbius decomposition - Born rule = Bergman projection (K67) - Reader exits with: substrate produces canonical QM observables

Chapter 8: Conservation Laws - Per-conservation-law theorems (SP-31-18, forthcoming Lyra) - Noether on substrate symmetries - Special attention: T (time reversal) and C (charge conjugation) currently NOT EXPLICIT - CPT from $SO_0(5,2)$ conformal structure - Information conservation via Reed-Solomon coding - Reader exits with: every conservation law derived, not postulated

Part IV — Substrate Uniqueness and Methodology

Chapter 9: Strong-Uniqueness Theorem - Current state v0.5 (10 criteria, null-model $(1/3)^{10} \approx 0.002\%$) - Per-criterion exposition: C1-C10 + C11-C14 candidates - Roadmap to v1.0 (curriculum-completion endpoint) - Reader exits with: precise statement of D_{IV}^{5s} ’s unique-forcing argument

Chapter 10: Methodology Stack - D/I/C/S tier system - Trichotomy: identifies / derives / predicts - F1-F4 Bridge Object family-member criteria - Mode 6 threshold formalization - K-audit chain governance (Cal/Keeper consensus + Casey Option

C) - Believability + Provability dual-axis evaluation - Reader exits with: how BST claims are vetted and tiered

Appendices

Appendix A: Casey-named principles - SWPP (Substrate Working Process Principle) - Five-Absence Predictions Set - Substrate Closure Principle - Graph Forces Principle - Integer Web Principle - Substrate Cognition Network Hypothesis (+ Cosmological Extension, DOUBLE-LOCKED EXTERNAL)

Appendix B: BST $\leftrightarrow$ Standard Physics Dictionary - Every BST term mapped to standard-physics equivalent (and back) - Example entries: | BST term | Standard term | |—| | substrate | “the vacuum / spacetime structure” | | primary integer | “fundamental constant” | | commitment cycle | “fundamental time unit / Planck cell” | | substrate-CHSH operator | “Bell correlation operator” | | Bergman projection | “Born rule” | | particle-as-winding | “particle as topological excitation” | | Bridge Object | “mathematical structure unifying multiple BST sources” | | Koonst tick | “sub-Planck substrate clock” | - Continuously expanded as Vol 1-10 introduce new terms

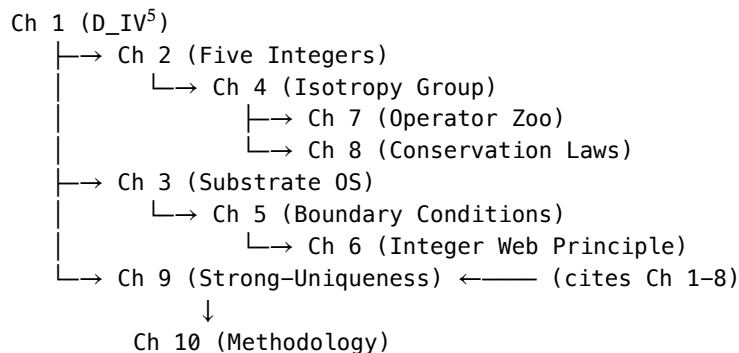
Theorem-numbering convention for curriculum

Existing: T1-T2427+ (AC graph theorems, sequential across all BST work)

Curriculum chapter cross-references: - Each curriculum chapter cites AC graph theorems with stable T-numbers - New theorems generated during curriculum work get T-numbers via /theorem claim - Vol 0 cites ~80-120 existing theorems + ~10-15 new (from SP-31-39 consolidations + chapter integration work)

Chapter section format: - Vol N, Chapter M, Section K, Subsection L (e.g., “Vol 0 Ch 7 Sec 2.3”) - T-references inline: “via T1925 (Why rank=2)” - AC graph paths shown for derivable claims: “T1 $\rightarrow$... $\rightarrow$ T1925 $\rightarrow$ claim”

Dependency tree (Vol 0 internal)



Reading order matters: Ch 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → 9 → 10. Forward references are minimized; backward references are heavy (Ch 9 cites everything before it).

Believability + Provability per-chapter rubric

Per Casey directive Wednesday EOD: dual-axis evaluation per chapter. A chapter is DONE only when BOTH axes pass.

Believability axis (passes when): - A physicist reads the chapter and recognizes the content - BST-to-standard-physics dictionary entries exist for all introduced terms - Examples in audience-familiar language (5th-grader test for motivation; PhD-physicist test for technical content) - Cal cold-read verdict: PASS or CONDITIONAL PASS

Provability axis (passes when): - All claims have D/I/C/S tier labels - All D-tier claims have AC graph theorem citations - All C-tier claims have explicit conditional structure - Keeper K-audit verdict: PASS

Status tracking per chapter: | Chapter | Believability | Provability | Overall | |—|—|—|—|
 |—| | Ch 1 | not started | not started | not started | | Ch 2 | partial (T1925, T1930 done; n_C, g pending SP-31-39) | partial | partial | | Ch 3 | partial (SWPP + Koons tick + 4-zone filed) | partial | partial | | Ch 4 | partial (operator zoo organizing principle just identified) | partial | partial | | Ch 5 | not started (SP-31-40 pending) | not started | not started | | Ch 6 | partial (Casey-named principle filed; integer-web mapping multi-week) | partial | partial | | Ch 7 | partial (5/11 operators in zoo; rest pending SP-31-6) | partial | partial | | Ch 8 | partial (most laws informal; C/T NOT EXPLICIT) | partial | partial | | Ch 9 | partial (v0.5 filed; v0.6+ pending) | partial | partial | | Ch 10 | partial (D/I/C/S + trichotomy + F1-F4 + Mode 6 all filed; M2C2 + Option C documented) | mostly done | mostly done |

Vol 0 v1.0 completion criteria

- All 10 chapters at Believability PASS + Provability PASS per rubric
- BST↔standard-physics dictionary complete for Vol 0 terms (~50-80 entries)
- All chapter dependencies cross-referenced
- Strong-Uniqueness Theorem at v0.6+ (per Lyra Task #206 extension)
- Cal cold-read of entire volume — PASS
- Keeper K-audit of entire volume — PASS

Estimated cadence: multi-month. Chapters 3, 4, 6, 10 mostly closeable in 4-6 weeks based on existing material. Chapters 5, 7, 8 gated on SP-31 sub-items (SP-31-1 Hilbert space + SP-31-6 operator zoo + SP-31-18 conservation laws + SP-31-40 BCs). Chapter 9 gated on Strong-Uniqueness v0.6+. Chapter 2 gated on SP-31-39 per-integer consolidation (quick).

Theorem-numbering convention extension proposal

For curriculum-internal theorems (generated DURING curriculum work, not pre-existing in AC graph): - Use prefix CT (Curriculum Theorem) followed by Vol-Ch-Th: e.g., CT0.2.3 = Vol 0 Ch 2 Theorem 3 - CT theorems get T-numbers via /theorem claim and AC graph integration - CT label is a curriculum-internal locator; T-number is the canonical AC graph identifier - Allows chapter-internal theorem-numbering for pedagogy without breaking AC graph hygiene

Casey: review CT proposal. If approved, adopt across all curriculum volumes.

Companion to Grace's Vol 0 chapter outline (forthcoming Thursday)

This scaffold + Grace's chapter outline = Vol 0 v0.2.

Grace's outline focuses on chapter CONTENT (what each chapter says, what existing material it absorbs). This scaffold focuses on chapter ARCHITECTURE (how chapters relate, theorem numbering, dictionary structure, completion rubric).

When both filed, Lyra absorbs into chapter-draft mode and starts producing chapter drafts. Cal cold-reads as drafts arrive. Keeper audits as drafts pass Cal. Multi-month cadence.

Status

Curriculum Vol 0 Architectural Scaffold v0.1 FILED Thursday 2026-05-21 morning. Companion to Grace's forthcoming chapter outline. Together = Vol 0 v0.2. 10 chapters + 2 appendices proposed. Theorem-numbering CT convention proposed (Casey review). Believability + Provability per-chapter rubric specified. Vol 0 v1.0 multi-month; gated on SP-31 sub-items + Strong-Uniqueness v0.6+.

— Keeper, 2026-05-21 Thursday 07:45 EDT